MID-TERM EXAMINATION
(CBCS)
Course Name: B. Tech (CSE/IT/CSEAI) , Semester: III
(Oct, 2022)

Subject Code: BAS-201 **Subject: Material Science & Engineering**
Time : 1 ½ Hours **Maximum Marks : 30**
Note: Q.1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) The critical field for niobium is 1×10^5 amp/m at 8 K and 2×10^5 amp/m at absolute zero. Find the transition temperature of the element.

(b) What are Type-I and type-II superconductors? (With Example)

(c) What is GMR (giant magneto resistance) effect?

(d) What is Fermi Energy? How it changes with temperature in an n-type semiconductor?

Q2 **(5,5)**

(a) A paramagnetic material has a magnetic field intensity of 10^4 Am^{-1}. If the susceptibility of the material at room temperature is 3.7×10^{-5}. Calculate the magnetization and flux density in the material.

(b) Mention four differences between the metal and ceramic materials with example?

Q3 **(5,5)**

(a) What is Spintronics? Why is it better than conventional electronics? Give some examples to explain how this technique works?

(b) What are the types of polarization in a dielectric material? How it varies with the frequency?

Q4 **(5,5)**

(a) What are the hard and soft magnetic materials? Explain the basic difference between them and the areas of their applications?

(b) A silicon plate of thickness 1 mm, breath 10mm and length 100mm is placed in a magnetic field of 0.5 Wb/m^2 acting perpendicular to its thickness. If 10^{-2} A current flows along its length, calculate the Hall voltage developed if the Hall coefficient is 3.66×10^{-4} m^3/coulomb.